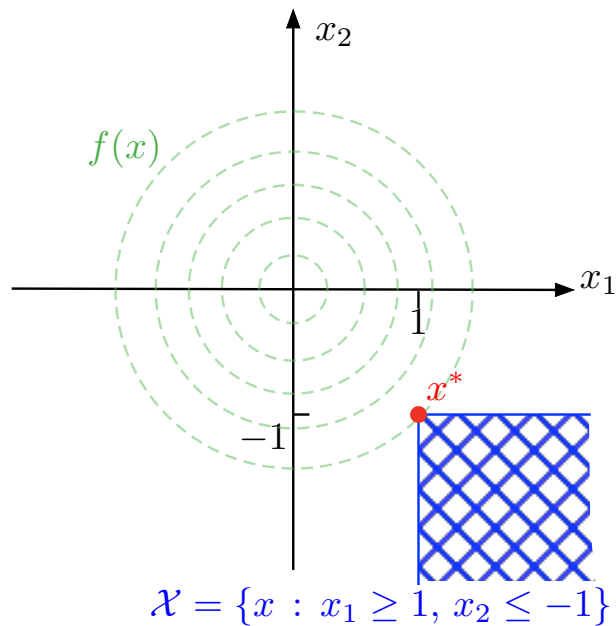


Solution 08
Soft Constraints and Robust MPC

Exercise 1 Soft constraints

- 1) $l_i(\epsilon_i) = u_i \epsilon_i$
 $u_i > \lambda_i^*$ for exactness, where λ_i is the optimal dual variable associated with the i th constraint
- 2) Graphically, the primal solution is $x^* = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$:



The Lagrangian is: $L(x, \lambda_1, \lambda_2) = x_1^2 + x_2^2 + \lambda_1(-x_1 + 1) + \lambda_2(x_2 + 1)$

Stationarity gives:

$$\nabla_x L = \begin{bmatrix} 2x_1 - \lambda_1 \\ 2x_2 + \lambda_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad \begin{aligned} \lambda_1^* &= 2x_1^* = 2 \\ \lambda_2^* &= -2x_2^* = 2 \end{aligned}$$

Furthermore, complementary slackness and dual feasibility are also satisfied.

Thus a valid choice for the penalty is:

$$l_i(\epsilon_i) = 3\epsilon_i \quad i = 1, 2$$

Exercise 2 Robust invariant set

$$\Omega_0 = [-10, 10]$$

$$\begin{aligned} \text{pre}^{\mathcal{W}}(\Omega_0) &= \{x \mid -10 \leq \frac{1}{2}x + w \leq 10 \text{ for } -1 \leq w \leq 1\} \\ &= \{x \mid -20 - 2w \leq x \leq 20 + 2w \text{ for } -1 \leq w \leq 1\} \\ &= \{x \mid -18 \leq x \leq 18\} \end{aligned}$$

$$\Omega_1 = [-10, 10] \cap [-18, 18] = [-10, 10] = \mathcal{O}_{\infty}^{\mathcal{W}}.$$